

to ask, What are the patterns of desire that are fed by these gestures? If I, as a novelist, choose to use brand names as elements in the depiction of character, do I not need to ask myself about the degree to which this makes me complicit in the manipulations of the marketplace?

In the same spirit, I think it also needs to be asked, What is it about climate change that the mention of it should lead to banishment from the preserves of serious fiction? And what does this tell us about culture writ large and its patterns of evasion?

In a substantially altered world, when sea-level rise has swallowed the Sundarbans and made cities like Kolkata, New York, and Bangkok¹ uninhabitable, when readers and museumgoers turn to the art and literature of our time, will they not look, first and most urgently, for traces and portents of the altered world of their inheritance? And when they fail to find them, what should they—what can they—do other than to conclude that ours was a time when most forms of art and literature were drawn into the modes of concealment that prevented people from recognizing the realities of their plight? Quite possibly, then, this era, which so congratulates itself on its self-awareness, will come to be known as the time of the Great Derangement.

2016

1. Capital of Thailand. "Kolkata": city in India.

ROBERT MACFARLANE

Born in 1976 in the village of Halam, Nottinghamshire, in England, Robert Macfarlane studied at the University of Cambridge, where he now teaches. Macfarlane is a distinguished nature writer whose nonfiction blends literary allusion, essayistic meditations, and novelistic plotlines in which individuals, like characters, experience revelations through their encounters with places. His early books, *Mountains of the Mind* (2003) and *The Wild Places* (2007), engage and challenge traditions of nature writing that align mountains with majesty and wilderness with empty spaces devoid of human intervention. Macfarlane approaches wildness as a "state of mind" rather than a feature of the land. His commitment to restoring human history and psychology to sites of geological and ecological wonder characterizes the skepticism of contemporary nature writing toward pristine notions of "Nature" with a capital N.

In his award-winning *Underland* (2019), Macfarlane uses the mythic motif of the underworld to connect the depths beneath the Earth's surface to the limits of human imagination and fortitude in the face of irreversible changes to the climate. Traversing caves beneath the Mendip Hills in England, underground nuclear waste facilities in New Mexico, and tunnels beneath the melting ice caps of Greenland, Macfarlane contemplates the paradox of human agency in the face of geological time. On the one hand, human beings have irrevocably changed the Earth's atmosphere and will leave the signature of our species in the strata of the Earth. On the other hand, humans are merely a blip in the deep time of our planet whose rock formations inscribe billions of years. In the excerpt included here, Macfarlane goes beneath Epping Forest in the northeast of London to explore the symbiotic networks between trees and fungus. Turning from megaflores to musrooms draws human attention to the aesthetically unremarkable forms of life that enable our

existence. Macfarlane takes the urban forest as a site of contradictions that leads its visitors to reappraise their own notions of preservation and extinction.

From UNDERLAND

The Understorey

(Epping Forest, London)¹

Occasionally—once or twice in a lifetime if you are lucky—you encounter an idea so powerful in its implications that it unsettles the ground you walk on.

The first time I heard anyone speak of the 'wood wide web,' more than a decade ago now, I was trying not to cry. A beloved friend was dying too young and too quickly. I had gone to see him for what I took to be the last time. He was tired by pain and drugs. We sat together, talked. My friend was a woodsman. Trees grew through his life and thought. His grandfather's surname was Wood, he lived in a timber-framed house that he had built himself, and he had planted thousands of trees by hand over the years. 'I have sap in my veins,' he wrote once.

That day I read aloud a poem that was important to us both, 'Birches' by Robert Frost,² in which climbing the snow-white trunks of birches becomes both a readying for death and a declaration of life. Then he told me about new research he had recently read concerning the interrelations of trees: how, when one of their number was sickening or under stress, they could share nutrients by means of an underground system that conjoined their roots beneath the soil, thereby sometimes nursing the sick tree back to health. It was a measure of my friend's generosity of spirit that—so close to death himself—he could speak unjealously of this phenomenon of healing.

He did not have the strength then to tell me the details of how this below-ground sharing operated—how tree might invisibly reach out to tree within the soil. But I could not forget the image of that mysterious buried network, joining single trees into forest communities. It was planted in my mind, and there took root. Over the years I would encounter other mentions of the same extraordinary idea, and gradually these isolated fragments began to connect together into something like understanding.

Merlin Sheldrake, as the oldest joke in mycology goes, is a fun guy to be around. During the days in which he conjures open the underland of Epping Forest for me, I ask more questions than I have of anyone for what feels like years. What he tells and shows me in that modest peri-urban forest reshapes my sense of the world in ways I am still processing.

1. Ancient woodland between London and Essex.

2. American poet (1874–1963).

'My childhood superheroes weren't Marvel characters,'³ Merlin once said to me, 'they were lichens and fungi. Fungi and lichen annihilate our categories of gender. They reshape our ideas of community and cooperation. They screw up our hereditary model of evolutionary descent. They utterly *liquidate* our notions of time. Lichens can crumble rocks into dust with terrifying acids. Fungi can exude massively powerful enzymes *outside* their bodies that dissolve soil. They're the biggest organisms in the world and among the oldest. They're world-makers and world-breakers. What's more superhero than *that*?'

Merlin and I set off on foot into Epping Forest one morning from a high clearing, heading roughly north, keeping the sun to the right of our line.

Epping extends to the north-east of London, and it is very far from a wild-wood. It was first designated as a royal hunting forest in the twelfth century by Henry II,⁴ with penalties for poaching that included imprisonment and mutilation. Presently it is managed by the City of London Corporation, and has more than fifty bye-laws governing behaviour within its bounds—though the punishments are now fiscal rather than corporal. It is fully contained within the M25, the orbital motorway that encircles outer London. Minor roads traverse it, and it is never more than two and a half miles wide. Despite its small extent, Epping is easy to get lost in—a forest of forking paths to which, for a thousand years, the people of London and its surrounds have gone for shelter, sex, escape and a relic greenwood magic.

Growl of roads. Whirr of a low-flying bumblebee, stirring the leaf litter with its downdraught. Buzzard overhead, turning, mewing. Old coppice trees left uncut, hydra-headed pollards. A fallen log, thick with moss; small orange fungi sprouting from wet breaks in its grain. Where trees thin out and light falls, hundreds of green beech seedlings are pushing up through the litter, none more than an inch high. Five fallow deer appear between hollies ahead of us, the dapple of leaf-light flicking off the dapple of their flanks as they move through the understorey.

In the language of forestry and forest ecology, the 'understorey' is the name given to the life that exists between the forest floor and the tree canopy: the fungi, mosses, lichens, bushes and saplings that thrive and compete in this mid-zone. Metaphorically, though, the 'understorey' is also the sum of the entangled, ever-growing narratives, histories, ideas and words that interweave to give a wood or forest its diverse life in culture.

'What interests me most,' says Merlin, 'is the understorey's understorey.' He points around at the beech, the hornbeam, the chestnut. 'All of these trees and bushes,' he says, 'are connected with one another below ground in ways we not only cannot see, but ways we have scarcely begun to understand.'

Merlin was taught as an undergraduate by Oliver Rackham,⁵ the legendary botanist whose research transformed our understanding of both the cultural and botanical history of the English landscape. Working with Rackham, Merlin found himself most intellectually attracted to places where orthodox evolutionary theory felt thinnest—and for him the thinnest

3. Brand of comics, with superhero characters including Spider-Man, Iron Man, and Captain America.

4. King of England (r. 1154–89).

5. Cambridge historian and ecologist (1939–2015).

places were where mutualisms were at work. Mutualism is a subset of symbiosis in which there exists between organisms a prolonged relationship that is interdependent and reciprocally beneficial.

'What fascinates me about mutualisms,' says Merlin, 'is that one would predict from basic evolutionary theory that they would be massively unstable, and collapse quickly into parasitism. But it turns out that there are very ancient mutualisms, which have remained stable for puzzlingly long times: between the yucca plant and yucca moths, for example, or of course between the bacteria that illuminate the bioluminescent headlamp of the bobtail squid, and the squid itself.'

'Of course,' I reply. 'The ancient glowing-bobtail-squid-and-bacteria mutualism.'

'The ultimate mutualism, though,' says Merlin, 'is between plants and mycorrhizal fungi.'

Merlin and I have been walking the forest for two hours or so when we reach one of Epping's great pollard beech groves. Pollarding—the pruning of the upper branches of a tree to promote dense growth—keeps trees alive for longer, indeed can enter them into an almost indefinite fairy-tale time of longevity. Here in the grove, long waving trunks yearn up to the sun. Through their leaves falls a green sub-sea light. It feels as if we are swimming through a kelp forest.

We stop and lie down for a while on the woodland floor, on our backs, not speaking, watching the trees' gentle movements in the breeze, and the light lacing and lancing from fifty feet or more above us. Where the pollards spread out to form the canopies, I realize I can trace patterns of space running along the edges of each tree's canopy: the beautiful phenomenon known as 'crown shyness', whereby individual forest trees respect each other's space, leaving slender running gaps between the end of one tree's outermost leaves and the start of another's.

Lying there among the trees, despite a learned wariness towards anthropomorphism, I find it hard not to imagine these arboreal relations in terms of tenderness, generosity and even love: the respectful distance of their shy crowns, the kissing branches that have pleached with one another, the unseen connections forged by root and hyphae between seemingly distant trees. I remember something Louis de Bernières⁶ has written about a relationship that endured into old age: 'we had roots that grew towards each other underground, and when all the pretty blossom had fallen from our branches we found that we were one tree and not two.' As someone lucky to live in a long love, I recognize that gradual growing-towards and subterranean intertwining; the things that do not need to be said between us, the unspoken communication which can sometimes tilt troublingly towards silence, and the sharing of both happiness and pain. I think of good love as something that roots, not rots, over time, and of the hyphae that are weaving through the ground below me, reaching out through the soil in search of mergings. Theirs, too, seems to me then a version of love's work.

Merlin gets up, walks towards the centre of the grove as if looking for something, then bends down and brushes away leaf litter and beech mast,⁷

6. English novelist (b. 1954).

7. Beech nuts fallen from a beech tree.

to clear a patch of soil the size of a saucer. I get up and follow him. He pinches some of the earth and rubs it between his fingers. It smears rather than crumbling: a rich, dark humus,⁸ made of composted leaves.

'This is our problem when it comes to studying the fungal network,' he says. 'Soil is fantastically impenetrable to experiments, and the fungal hyphae are on the whole too thin to see with the naked eye. That's the main reason it's taken us so long to work out the wood wide web's existence, and to discern what it's doing.'

Rivers of sap flow in the trees around us. If we were right now to lay a stethoscope to the bark of a birch or beech, we would hear the sap bubbling and crackling as it moves through the trunk.

'You can put rhizotrons⁹ into the ground to look at root growth,' Merlin says, 'but those don't really give you the fungi because they're too fine. You can do below-ground laser scanning but, again, that's too crude for the fungal networks.'

I am reminded once more of how resistant the underland remains to our usual forms of seeing; how it still hides so much from us, even in our age of hyper-visibility and ultra-scrutiny. Just a few inches of soil is enough to keep startling secrets, hold astonishing cargo: an eighth of the world's total biomass comprises bacteria that live below ground, and a further quarter is of fungal origin.

'We know the network is there,' Merlin says, 'but it's so effortful to track it. So we have to look for clues to the labyrinth—find clever means of following its paths.'

I kneel beside him. I can see dozens of insects just in this small area, the names of most of which I do not know: gleaming spiders and red-bronze beetles battling over the leaves, a woodlouse curled up into a sphere, a green threadworm writhing through the humus.

'It's *roiling* with life,' I say to Merlin.

'That's just the visible life. Hyphae will be growing into the decomposing matter of this half-rotting leaf,' says Merlin, 'into those rotting logs and those rotting twigs, and then you'll have the mycorrhizal fungi whose hyphae grow into hot spots—all of them frothing and tangling and fusing, making a network that's connecting holly to holly but also to this beech, and to a seedling of something else over there, layering and layering and layering—until, well, it blows your computational brain!'

As Merlin speaks I feel a quick, eerie sense of the world shifting irreversibly around me. Ground shivering beneath feet, knees, skin. *If only your mind were a slightly greener thing, we'd drown you in meaning* . . . I glance down, try to trance the soil into transparency such that I can see its hidden infrastructure: millions of fungal skeins suspended between tapering tree roots, their prolific liaisons creating a gossamer web at least as intricate as the cables and fibres that hang beneath our cities. What's the haunting phrase I've heard used to describe the realm of fungi? *The kingdom of the grey*. It speaks of fungi's utter otherness—the challenges they issue to our usual models of time, space and species.

'You look at the network,' says Merlin, 'and then it starts to look back at you.'

8. Organic component of soil.

9. Underground labs made to study soil.

Potawatomi, a Native American language of the Great Plains region, includes the word *puhpowee*, which might be translated as 'the force which causes mushrooms to push up from the earth overnight'. In 'all its technical vocabulary', Robin Wall Kimmerer notes, 'Western science has no such term, no words to hold this mystery.'

Kimmerer herself is a member of the Citizen Potawatomi Nation. A speaker of what she calls 'fluent botany', she is careful to distinguish this from what she refers to as 'the language of plants'—that is to say, the language that plants speak, as opposed to the language that is used to speak of plants. Kimmerer does not disdain the precisions of botanical lexis,¹ which 'polishes the gift of seeing', but she finds it also to be of necessity a lexis of objectification and distancing, with something missing beneath its finely faceted surface. That missing something is predominantly the acknowledgement of life in the more-than-human world, an indifference which is grained into language not just at the level of individual words, but at the deeper-down levels of grammar and syntax.

In Potawatomi, by contrast, almost all words declare the animacy or inanimacy of that to which they refer. The language is predisposed to recognize life in otherness, and also to extend the reach of that category of 'life' far beyond its familiar limits in Western thought. In Potawatomi, not only humans, animals and trees are alive, but so too are mountains, boulders, winds and fire. Stories, songs and rhythms are all also animate, they *are*, they *be*. Potawatomi is a language abundant with verbs: 70 per cent of its words are verbs, compared to 30 per cent in English. *Wiikwegamaa*, for instance, means 'to be a bay'. 'A bay is a noun only if water is *dead*,' writes Kimmerer:

trapped between its shores and contained by the word. But the verb . . . releases the water from bondage and lets it live. 'To be a bay' holds the wonder that, for this moment, the living water has decided to shelter itself between these shores, conversing with cedar roots and a flock of baby mergansers.²

Like Kimmerer, I wish for a language that recognizes and advances the animacy of the world, 'the life that pulses through pines and nuthatches and mushrooms . . . well[ing] up all around us'. Like Kimmerer, I relish those aspects of discourse that extend being and sentience respectfully and flexibly beyond the usual bearers of such qualities. Like Kimmerer I believe that we need, now, a 'grammar of animacy'. A modern predisposition to regard animacy as anomaly runs through what the poet Jeremy Prynne³ once called 'mammal language', by which he means the language that is used by humans, encoding intent, agency and muscular power deep in its grammar.

The real underland of language is not the roots of single words, but rather the soil of grammar and syntax, where habits of speech and therefore also habits of thought settle and interact over long periods of time. Grammar and syntax exert powerful influence on the proceedings of language and its users. They shape the ways we relate to each other and to the living world. Words are world-makers—and language is one of the great geological forces of the Anthropocene.

1. The vocabulary of a language.

2. From Robin Wall Kimmerer, *Braiding Sweetgrass: Indigenous Wisdom, Scientific Knowledge,*

and the Teachings of Plants (Minneapolis: Milkweed, 2013).

3. English poet (b. 1936).

Projects have recently been started around the world to gain even the most basic of vocabularies for the experiences of life and death in the Anthropocene. These stuttering attempts to speak what it is we are doing have generated ugly new terms for an ugly epoch: 'geotraumatism', 'planetary dysphoria', 'apex-guilt'. Such words feel like futile forms of nominalism, a hopelessly hyperactive pointing and naming. They stick in the throat in two ways: they are difficult to utter and hard to swallow.

Only one of these recent coinages resonates with me: 'species loneliness', for the intense solitude that we are fashioning for ourselves as we strip the Earth of the other life with which we share it. If there is human meaning to be made of the wood wide web, it is surely that what might save us as we move forwards into the precarious, unsettled centuries ahead is collaboration: mutualism, symbiosis, the inclusive human work of collective decision-making extended to more-than-human communities.

You look at the network, and then it starts to look back at you . . .

Writing of mycorrhizal fungi, Albrecht proposes that we rechristen the Anthropocene, naming it instead the Symbiocene—an epoch characterized in terms of social organization 'by human intelligence that replicates the symbiotic and mutually reinforcing life-reproducing forms and processes found in living systems . . . such as the wood wide web.'

The word for world is forest.

2019

POETRY OF CLIMATE CHANGE

In the twenty-first century, oceans are heating up, ice sheets are shrinking, glaciers are melting, sea levels are rising, biodiversity is falling, and extreme weather is battering the planet. Although environmental awareness can be traced through the Romanics to earlier poets like Spenser and Chaucer, it was not until the twenty-first century that poets began grappling with the consequences of anthropogenic (human-caused) climate change. Their poems take a variety of approaches, but all have forged innovative ways of seeing, responding to, and thinking about the fate of the planet.

Some, such as Simon Armitage's "Ark," are openly elegiac, mourning the loss of diverse species, the degradation of reefs, ice sheets, and snow caps. Others, such as Peter Reading's "Clockwise (from the bottom)," assume a prophetic stance, satirically foretelling devastation. Still others, such as Vahni Capildeo's "Book of Dreams / Livre de Cauchemars," VI, present a dream vision of the future—or nightmare, as the French subtitle has it. In some climate poems, such as Reading's, the perspective is global. In others, such as Capildeo's poem set in the Caribbean, the focus is on the local effects of worldwide warming. Similarly, Seamus Heaney scrutinizes a glacier melting in Iceland, Patience Agbabi overhears rivers flooding in climate-sensitive Nepal, and in "Rainforest in the Sleep Room," Pascale Petit recounts the deforestation and bio-depletion of the Amazon, a human intrusion she figuratively compares to a lobotomy.

Climate change poses difficulties for representing time and space. How can a brief poem convey the long, layered timescale of climatic shift? In the compressed language of "Höfn," Heaney evokes the ages-long processes of glacier formation, followed by rapid, fearful warming. In "ECO,nomics" Agbabi reflects another temporal complexity of climate change, how seasons become confused, as winter merges with spring. Or as Craig Santos Perez wryly puts it, "It was summer all winter long." Like time, space becomes confused. Recalling a long-haul flight from